

AI-enhanced Flipped CLIL Models for ESP Education: A Systematic Review in Tourism and Hospitality Contexts

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Abstract

This study presents a systematic review of empirical research examining the integration of Artificial Intelligence (AI), flipped classroom strategies, and Content and Language Integrated Learning (CLIL) in English for Specific Purposes (ESP) instruction, with a particular focus on tourism and hospitality contexts. Guided by the PRISMA 2020 guidelines and the PSALSAR framework, peer-reviewed journal articles published between 2016 and 2024 were systematically identified, screened, and synthesized from major academic databases. A total of 18 empirical studies met the inclusion criteria and were included in the final analysis. The findings indicate that ESP instruction grounded in needs-based pedagogy is most effectively supported when AI, flipped learning, and CLIL operate as complementary pedagogical components. AI primarily functions as a pedagogical enabler by supporting personalization, formative feedback, and learner autonomy, while flipped classroom strategies restructure learning activities toward interactive and task-based engagement. CLIL further facilitates the integration of language development with domain-specific knowledge essential for professional communication. Notably, the synthesis suggests that these approaches collectively form an instructional ecology, in which instructional relevance, learning activities, and technological support are interdependently aligned. Despite promising outcomes, the review reveals a limited number of studies that holistically integrate all four dimensions. This study, therefore, provides a consolidated pedagogical perspective and identifies directions for future research on AI-enhanced flipped CLIL models in ESP education. This review contributes by conceptualizing AI-enhanced flipped CLIL in ESP education as an instructional ecology, in which Artificial Intelligence, flipped classroom pedagogy, and CLIL function as interdependent pedagogical mechanisms rather than isolated techniques.

Keywords: systematic review, flipped classroom, artificial intelligence in education, English for Specific Purposes, Content and Language Integrated Learning

1. Introduction

The tourism and hospitality industry is a globally interconnected sector that requires professionals to communicate effectively in English within diverse, dynamic, and culturally situated contexts. Unlike general English instruction, English for Specific Purposes (ESP) in this domain emphasizes profession-specific communication, situational language use, and task-oriented interaction aligned with real workplace demands (Dudley-Evans & St John, 1998; Richards & Rodgers, 2014). However, existing ESP instruction has often been criticized for relying on decontextualized language practice and teacher-centered approaches that insufficiently prepare learners for authentic professional communication.

To address these challenges, Content and Language Integrated Learning (CLIL) has been increasingly adopted as a pedagogical approach that enables the simultaneous development of language proficiency and domain-specific knowledge (Coyle et al., 2010; Dalton-Puffer, 2011). In tourism and hospitality education, CLIL has demonstrated potential to enhance contextualized learning by embedding language instruction within discipline-relevant content and professional scenarios. At the same time, flipped classroom strategies restructure instructional time by relocating content delivery to pre-class phases and prioritizing interactive, task-based learning during class sessions, which aligns well with ESP and CLIL pedagogical principles (Bergmann & Sams,

2012; Simonova et al., 2021; Liu et al., 2022). Prior studies suggest that flipped learning aligns well with ESP and CLIL principles by promoting learner autonomy, collaboration, and applied language use.

More recently, advances in Artificial Intelligence (AI), including adaptive learning systems, chatbots, intelligent tutoring systems, and automated feedback tools, have created new opportunities to enhance language instruction. In ESP contexts, AI has been used to support personalized learning, formative assessment, and self-regulated learning processes (Luckin et al., 2016; Holmes et al., 2019). Despite growing interest, existing research tends to examine AI-supported instruction, flipped classrooms, or CLIL in isolation, with limited attention to their combined pedagogical potential, particularly in tourism and hospitality education.

Given this fragmented research landscape, a systematic synthesis of empirical studies integrating these approaches is needed (Page et al., 2021; Rapti & Panagiotidis, 2024). While several reviews have explored AI in language learning or flipped classroom applications in ESP, no systematic review has comprehensively examined how Artificial Intelligence, flipped classroom strategies, and CLIL are jointly implemented within ESP instruction in professional tourism and hospitality contexts. Addressing this gap, the present study conducts a systematic review of empirical research published between 2016 and 2024 to synthesize existing evidence on instructional designs, pedagogical roles of AI, learning activities, assessment practices, and reported outcomes.

Guided by the PRISMA 2020 guidelines and the PSALSAR framework, this systematic review aims to (1) examine how flipped classroom strategies are applied in ESP and CLIL contexts, (2) identify the pedagogical roles of AI in flipped or CLIL-oriented language instruction, (3) analyze reported benefits and limitations of AI integration in ESP classrooms, and (4) highlight research gaps that inform future empirical studies. By consolidating current evidence, this review seeks to provide a coherent pedagogical perspective and a conceptual foundation for advancing research on AI-enhanced flipped CLIL approaches in ESP education.

2. Theoretical Framework

This systematic review is underpinned by an integrated theoretical framework that brings together English for Specific Purposes (ESP), Content and Language Integrated Learning (CLIL), and technology-enhanced learning, with Artificial Intelligence (AI) and flipped classroom strategies positioned as pedagogical enablers rather than standalone theoretical models. The framework serves as a conceptual lens for synthesizing how instructional design, learning activities, and assessment practices are structured in contemporary ESP research, particularly within professional domains such as tourism and hospitality (Dudley-Evans & St John, 1998; Coyle et al., 2010; Holmes et al., 2019).

2.1 English for Specific Purposes (ESP) as the Core Pedagogical Foundation

English for Specific Purposes (ESP) constitutes the primary pedagogical foundation of this review (Dudley-Evans & St John, 1998; Anthony, 2018). ESP instruction prioritizes functional communication, domain-specific vocabulary, and authentic language use within professional contexts. In tourism and hospitality education, ESP defines the communicative goals, task relevance, and contextual appropriateness of instructional practices, thereby providing the central criterion for evaluating the effectiveness of AI-supported and flipped learning approaches (Anthony, 2018; Pham & Phuong, 2024).

2.2 Content and Language Integrated Learning (CLIL) as an Instructional Integration Mechanism

Content and Language Integrated Learning (CLIL) provides the instructional mechanism through which language learning and subject-specific knowledge are simultaneously developed (Coyle et al., 2010). CLIL supports the integration of cognitive, linguistic, and professional competencies, enabling learners to engage with authentic disciplinary content while developing communicative proficiency (Coyle et al., 2010; Dalton-Puffer, 2011). Within ESP contexts, particularly in tourism and hospitality, CLIL facilitates contextualized learning by aligning language instruction with real-world professional practices and situational communication demands (Rodrigues et al., 2022).

2.3 Flipped Classroom as a Pedagogical Structuring Strategy

The flipped classroom model is conceptualized as a pedagogical structuring strategy that reallocates instructional time toward active, task-based engagement (Bergmann & Sams, 2012; Simonova et al., 2021; Liu et al., 2022). By relocating content delivery to pre-class phases and reserving in-class time for interaction, collaboration, and problem-based activities, flipped learning promotes active engagement and learner responsibility (Bergmann & Sams, 2012; Simonova et al., 2021; Liu et al., 2022). In ESP and CLIL settings, flipped classroom strategies support task-based interaction and professional language practice, allowing learners to apply domain-specific knowledge through communicative activities aligned with workplace scenarios (Liu et al., 2022).

2.4 Artificial Intelligence as a Pedagogical Enabler in ESP Instruction

Artificial Intelligence is positioned as a pedagogical enabler that enhances the implementation of ESP, CLIL, and flipped classroom approaches. Rather than functioning as an independent instructional theory, AI supports adaptive learning, personalized feedback, learner autonomy, and formative assessment through technologies such as chatbots, intelligent tutoring systems, adaptive platforms, and automated feedback tools (Holmes et al., 2019; Luckin et al., 2016; Yang, 2022). Prior studies suggest that AI-supported instruction strengthens self-regulated learning and contextualized language practice without replacing pedagogical decision-making by instructors (Luckin et al., 2016; Yang, 2022).

2.5 Integrated Framework for Systematic Synthesis

Collectively, ESP defines what learners need to learn, CLIL determines how language and professional content are integrated, flipped classroom strategies organize when and where learning activities occur, and AI enhances how effectively instruction, feedback, and assessment are delivered. This integrated theoretical framework guided the systematic synthesis by enabling the identification of recurring pedagogical patterns, instructional roles of AI, and research gaps across the reviewed studies. It provides a coherent conceptual foundation for interpreting current practices and for informing future research aimed at developing and validating comprehensive AI-enhanced flipped CLIL models in ESP education (Coyle et al., 2010; Holmes et al., 2019; Rapti & Panagiotidis, 2024). This integrated theoretical framework guided the systematic synthesis by enabling the identification of recurring pedagogical patterns and instructional roles of AI across the reviewed studies. This framework also serves as an analytical lens for coding, categorizing, and synthesizing empirical patterns across the reviewed studies.

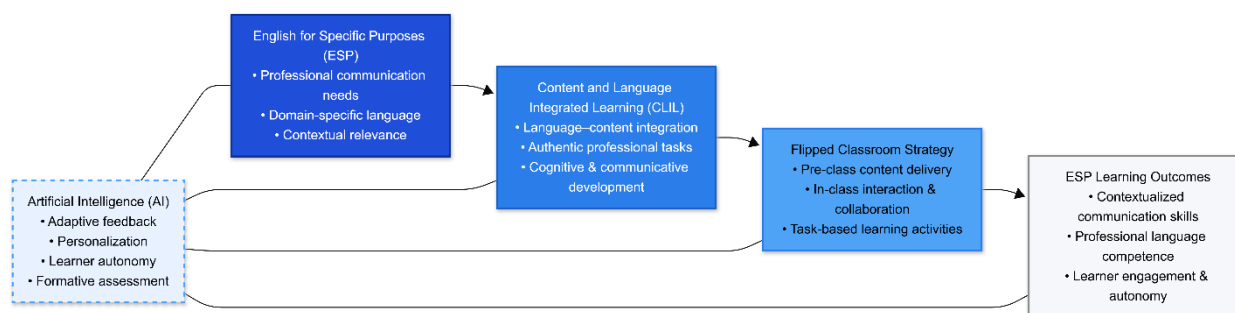


Figure 1. Conceptual Framework of AI-Enhanced Flipped CLIL for ESP Education

Figure 1 illustrates the conceptual framework underpinning this systematic review. derived from synthesis. English for Specific Purposes (ESP) serves as the core pedagogical foundation, defining professional language objectives and contextual relevance. Content and Language Integrated Learning (CLIL) functions as the instructional integration mechanism, enabling the simultaneous development of language proficiency and domain-specific knowledge. Flipped classroom strategies structure learning across pre-class and in-class phases, promoting active, task-based engagement. Artificial Intelligence (AI) operates as a pedagogical enabler that enhances personalization, formative feedback, learner autonomy, and assessment processes. Together, these components interact to support contextualized communication skills and professional competence in tourism and hospitality education. This is not a proposed model but a synthesized conceptual framework.

3. Research Methodology

This study adopted a systematic literature review (SLR) design to synthesize empirical evidence on the integration of Artificial Intelligence (AI), flipped classroom strategies, Content and Language Integrated Learning (CLIL), and assessment practices in English for Specific Purposes (ESP) instruction. The review was conducted following the PSALSAR framework (Protocol, Search, Appraisal, Synthesis, Analysis, and Report) and aligned with the PRISMA 2020 guidelines, ensuring methodological rigor, transparency, and replicability (Page et al., 2021).

During the protocol stage, the scope of the review was clearly defined to focus on empirical, classroom-based studies that examined the application of AI, flipped learning, and/or CLIL within ESP contexts, particularly in professional and vocational domains such as tourism and hospitality. Explicit research questions guided the review process and structured the subsequent synthesis.

In the search stage, a systematic keyword-based search was conducted across three major academic databases—Scopus, Web of Science (WoS), and ERIC—which are widely recognized as authoritative sources for peer-reviewed research in education and applied linguistics (Page et al., 2021). The search covered peer-reviewed journal articles published between 2016 and 2024. Search strings combined key terms related to flipped classrooms, artificial intelligence, CLIL, and ESP to ensure comprehensive coverage of relevant literature. Table 1 summarizes the core keywords and derivative search terms used to identify relevant journal articles across the selected databases. Table 2 presents the number of journal articles identified through keyword searches across ERIC, Web of Science, and Scopus prior to screening and duplicate removal.

The appraisal stage involved a two-step screening process. First, titles and abstracts were screened based on predefined inclusion and exclusion criteria. Second, full-text articles were assessed using quality assessment criteria focusing on empirical rigor, clarity of instructional design, relevance to ESP contexts, and completeness of methodological reporting. Only empirical, classroom-based studies published between 2016 and 2024 were included, in line with established systematic review practices (Page et al., 2021). Some studies meeting all quality thresholds were retained for synthesis. The inclusion and exclusion criteria applied during the screening process are summarised in Table 3.

The synthesis and analysis stages employed a qualitative systematic synthesis approach. This review does not aim to conduct a statistical meta-analysis due to substantial heterogeneity in study designs, learning outcomes, instructional contexts, and AI tools across the included studies. Data were extracted and organized into comparative tables to identify recurring instructional patterns, pedagogical roles of AI, learning activities in flipped and CLIL environments, assessment practices, and reported outcomes. Rather than conducting statistical meta-analysis or primary quantitative, the study emphasized thematic categorization, cross-study comparison, and pattern identification across the selected literature. The quality assessment criteria applied at the eligibility stage are summarised in Table 4.

Finally, the reporting stage documented the entire review process in accordance with PRISMA standards, including the presentation of search results, screening decisions, and synthesized findings. This systematic synthesis provides a robust evidence base for understanding current pedagogical practices and research gaps, and it serves as a foundation for the future development of AI-enhanced flipped CLIL instructional models in ESP education.

Table 1. Keywords and Their Derivatives in Journal Article Searches

Item.	Main Keywords	Derivative Keywords
1.	Flipped Classroom and ESP	1. Active learning 2. Student-centered learning 3. Flipped learning 4. Technology-enhanced instruction 5. In-class collaborative activities 6. Inverted classroom
2.	CLIL AND ("Artificial Intelligence" OR "AI")	1. AI-enhanced CLIL 2. AI-supported bilingual education 3. Technology-assisted CLIL instruction 4. AI in content and language integration
3.	Flipped classroom AND ("Artificial Intelligence" OR "AI")	1. AI-enhanced flipped classroom 2. Intelligent flipped learning environment 3. Adaptive flipped learning 4. AI-powered flipped instruction 5. Personalized flipped classroom via AI 6. Flipped learning with generative AI 7. Flipped learning with automated feedback 8. Flipped learning enhanced by NLP tools
4.	English for Specific Purposes AND ("Artificial Intelligence" OR "AI")	1. AI-enhanced ESP instruction 2. AI for English for Occupational Purposes 3. ESP with AI-assisted learning 4. ESP content generation with generative AI

Table 2. Number of Journal Articles Identified through Keyword Searches across Databases

Item	SLR Keywords	Database	Results
1	"Flipped classroom" AND "English for Specific Purposes"	ERIC	73
		WoS	17
		Scopus	16
2	"CLIL" AND ("Artificial Intelligence" OR "AI")"	ERIC	1
		WoS	9
		Scopus	18
3	"Flipped classroom" AND ("Artificial Intelligence" OR "AI")"	ERIC	49
		WoS	82
		Scopus	320
4	"English for Specific Purposes" AND ("Artificial Intelligence" OR "AI")"	ERIC	5
		WoS	30
		Scopus	43

Note. ERIC = Education Resources Information Center; WoS = Web of Science; Scopus = Elsevier database.

Table 3. Inclusion and Exclusion Criteria

Item	Inclusion Criteria	Exclusion Criteria
1	Empirical journal articles focusing on ESP instruction	Literature reviews, conceptual papers, books, websites
2	Studies integrating AI, Flipped Classroom, and/or CLIL	Studies unrelated to AI, Flipped Classroom, or CLIL
3	Articles published between 2016-2024	Publications before 2016
4	English-language publications	Non-English publications
5	Full-text accessible journal articles	Abstract-only or inaccessible full texts
6	Articles indexed in Scopus, Web of Science, or ERIC	Duplicate records

Note. Screening criteria were applied during the identification and screening phases of the PRISMA process.

Table 4. Quality Assessment Criteria (Eligibility Stage)

Item	Inclusion Criteria	Exclusion Criteria
1	Empirical, classroom-based, or intervention-based studies	Opinion papers, dissertations, conference proceedings
2	Clear description of participants, context, and procedures	Insufficient methodological detail
3	Explicit instructional design or intervention description	Vague or unspecified instructional implementation
4	Reported outcomes related to ESP learning or teaching	Outcomes unrelated to ESP contexts
5	Conducted in educational or ESP professional contexts	Studies outside educational or ESP-related settings

Note. Studies excluded at this stage did not meet minimum methodological quality requirements.

Table 5 outlines the rationale underpinning the inclusion and exclusion criteria applied in this systematic review to ensure relevance, methodological rigour, and alignment with the study objectives.

Table 5. Justification for Inclusion and Exclusion Criteria

Item	Criteria	Rationale
1	Topic relevance: Journal articles on the use of AI with the flipped classroom technique for ESP classroom	Ensures that selected studies directly address the integration of AI and the flipped classroom approach within ESP contexts, maintaining conceptual focus and relevance to the study objectives.
2	Databases: Scopus, Web of Science, and ERIC	These databases are reputable and comprehensive sources of peer-reviewed research in education and technology. Restricting the search to these enhances academic credibility and replicability.
3	Publication years: 2016-2024	Limits the review to recent and relevant studies that reflect current advancements in AI and flipped learning. Earlier studies may not align with modern technological applications or pedagogical developments.
4	Language: English-language journal articles only	Ensures consistency in analysis and interpretation. Excluding non-English studies prevents translation bias and contextual inconsistencies.
5	Accessibility: Only full-text, accessible studies included	Guarantees that all reviewed studies can be critically evaluated in detail. Excluding inaccessible or abstract-only papers prevents partial or biased analysis.
6	Publication type: Empirical journal articles only	Focuses the review on data-driven research with measurable outcomes, excluding literature reviews, books, websites, and conference papers to maintain methodological rigor.
7	Duplication: Removal of repeated records across databases	Prevents redundancy and ensures each study contributes unique findings to the systematic synthesis.

Note. These criteria were applied during the eligibility stage of the PRISMA process to ensure relevance, methodological rigour, and alignment with the study objectives.

Following the initial eligibility screening, a quality assessment (QA) process was conducted to further evaluate the methodological rigor and contextual relevance of the selected studies. Articles were excluded if they lacked sufficient methodological detail or relevance to ESP contexts, following PRISMA-recommended appraisal procedures (Page et al., 2021). This two-stage selection process aligns with PRISMA recommendations and enhances transparency in the study selection procedure. The specific reasons for exclusion at the QA stage are summarized in Table 6.

Table 6. Reasons for Exclusion Based on Quality Assessment (QA)

QA Code	Exclusion Reason	Description
QA1	Not ESP context	The study did not focus on English for Specific Purposes instruction or professional language use.
QA2	Not empirical research	The study was theoretical, review-based, or conceptual, and did not include empirical data or classroom intervention.
QA3	No AI integration	The study did not involve the use of artificial intelligence tools or AI-supported instructional practices.
QA4	No CLIL component	The study did not incorporate content and language integrated learning principles or practices.
QA5	Non-classroom-based	The study was not conducted in a classroom or instructional setting.
QA6	Incomplete methodological reporting	The research design, procedures, participants, or data analysis were insufficiently described to allow critical appraisal.

Data Appraisal and Analysis Procedures

The inspection stage of appraisal encompassed quality appraisal and literature evaluation. Rechecking the inclusion and exclusion criteria formed an essential part of the literature appraisal process to ensure consistency and relevance. The quality assessment consisted of two key evaluation questions:

- QA1: Does the article focus on English for Specific Purposes (ESP) instruction, particularly within professional or vocational contexts such as tourism and hospitality?
- QA2: Does the article involve empirical, classroom-based research (e.g., action research, quasi-experiments, or teaching interventions)?

The synthesis stage involved systematically extracting and categorizing data from the selected journal articles. Extracted information was organized into comparative tables, enabling the identification of similarities and differences across studies. The synthesis focused on major analytical themes, including AI integration, flipped classroom strategies, CLIL implementation, and assessment tools employed in ESP instruction.

Subsequently, the analysis stage focused on interpreting the synthesized data to identify recurring patterns, reported findings, and existing research gaps. This stage also addressed the research questions by evaluating the relevance, applicability, and strength of the evidence collected.

Finally, the reporting stage documented all procedures undertaken throughout the review process and presented the findings in accordance with academic publication standards. The outcomes of this stage provided a foundation for developing and validating the proposed instructional model.

4. Results

The PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework was employed to ensure transparency, rigor, and reproducibility in the article selection process for this systematic literature review (Page et al., 2021). The process consisted of four main stages: Identification, Screening, Eligibility, and Inclusion, each documented in a flow diagram.

To explore current research on innovative approaches in English for Specific Purposes (ESP), four separate keyword searches were conducted:

1. "flipped classroom" AND "English for Specific Purposes"
2. "CLIL" AND ("Artificial Intelligence" OR "AI")
3. "flipped classroom" AND ("Artificial Intelligence" OR "AI")
4. "English for Specific Purposes" AND ("Artificial Intelligence" OR "AI")

These searches were performed across academic databases including ERIC, WoS, and Scopus between 2016 and 2024. The goal was to identify peer-reviewed journal articles that discuss empirical

ESP instruction incorporating technology-enhanced methodologies.

PRISMA Stages Defined:

- Identification: All potentially relevant articles were collected using predefined keywords. This resulted in a large pool of records for each search.
- Screening: Duplicate entries were removed, and titles and abstracts were examined for relevance to ESP instruction and educational innovation.
- Eligibility: Full-text articles were reviewed to determine whether they met the inclusion criteria (e.g., focus on ESP, use of flipped classroom, AI, or CLIL, published in English, between 2016-2024).
- Inclusion: Articles that met all criteria were selected for data synthesis and further analysis. These selected studies formed the evidence base for designing the AIEFC instructional model.

Each figure (Figures 2-5) in the study presents a separate PRISMA diagram corresponding to the search results for one keyword pair. The diagrams help visualize the number of articles identified, filtered, and retained at each stage of the review process, ensuring the methodological transparency of the study.

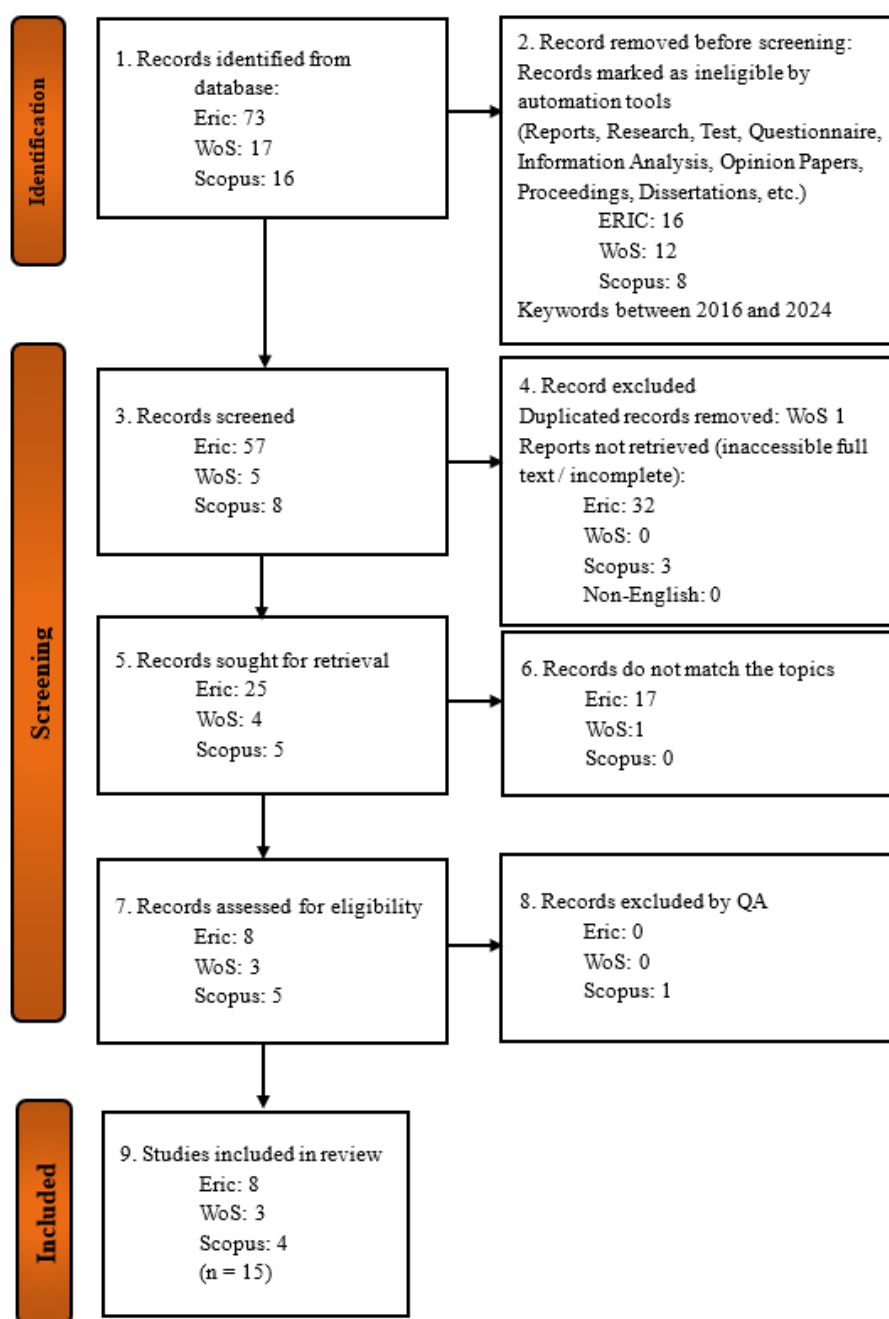


Figure 2. Result of Article Search ("flipped classroom" AND "English for Specific Purposes")

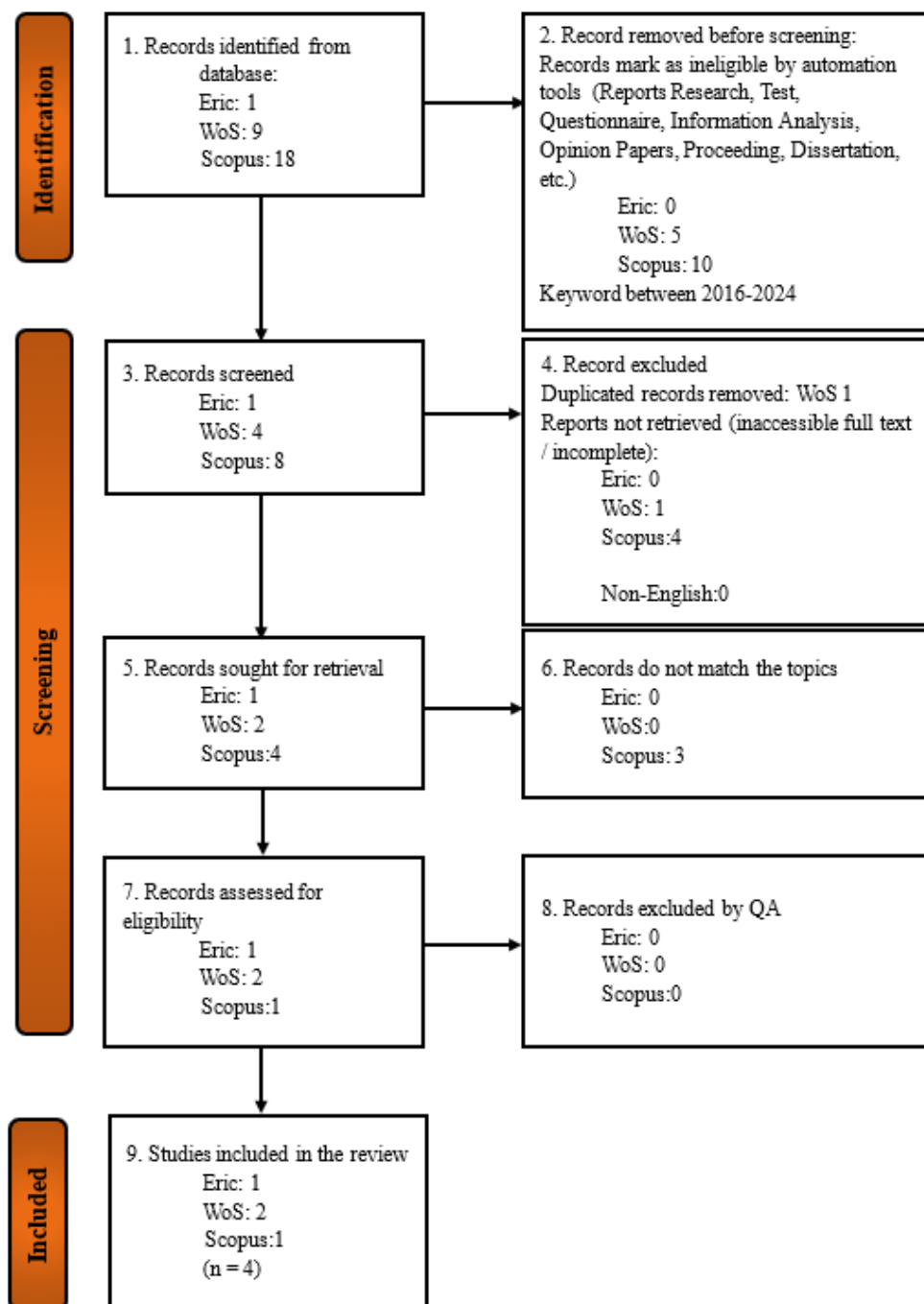


Figure 3. Result of Article Search "cil" AND ("artificial intelligence" OR "ai")

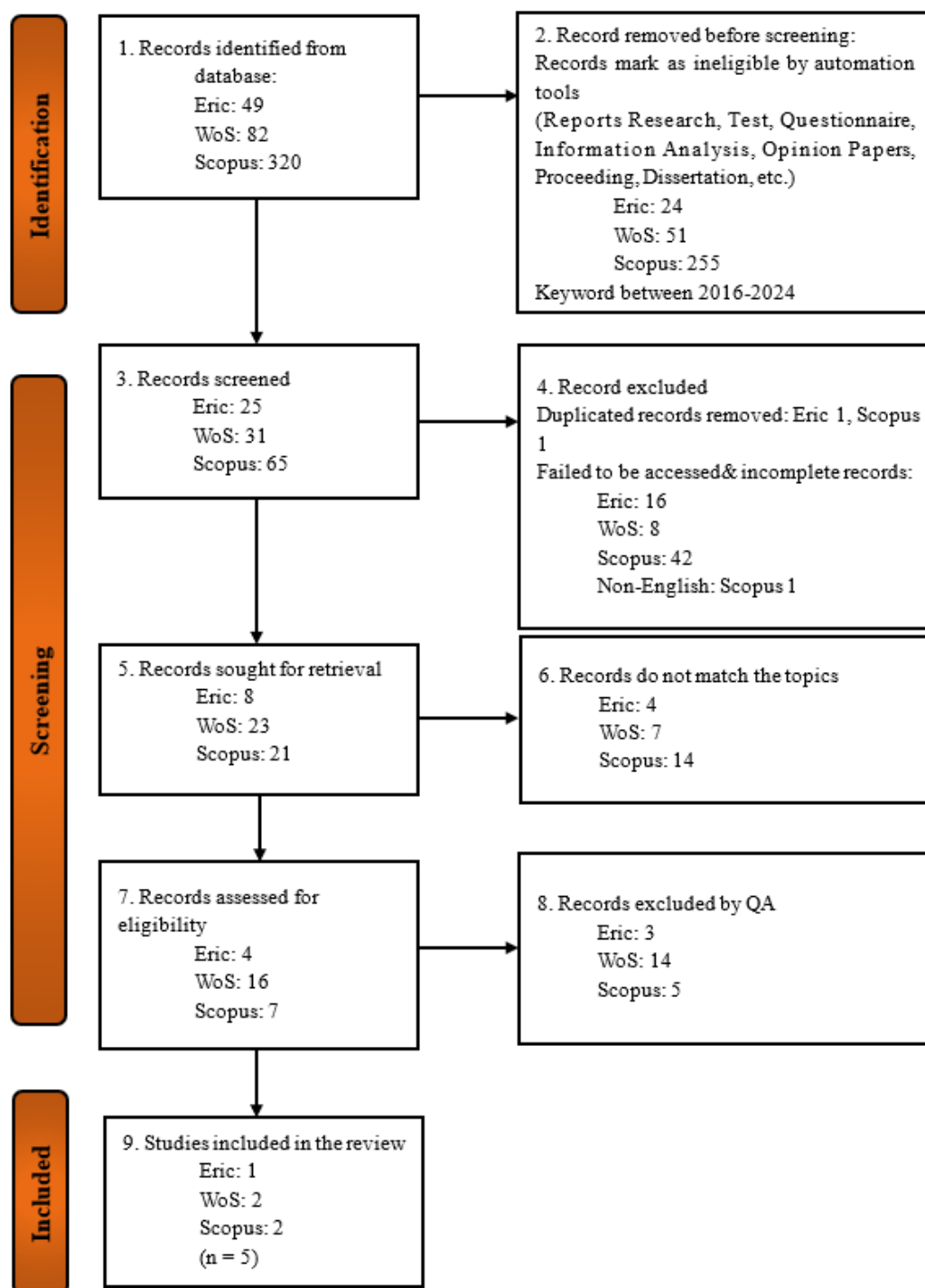


Figure 4. Result of Article Search "flipped classroom" AND ("artificial intelligence" OR "ai")

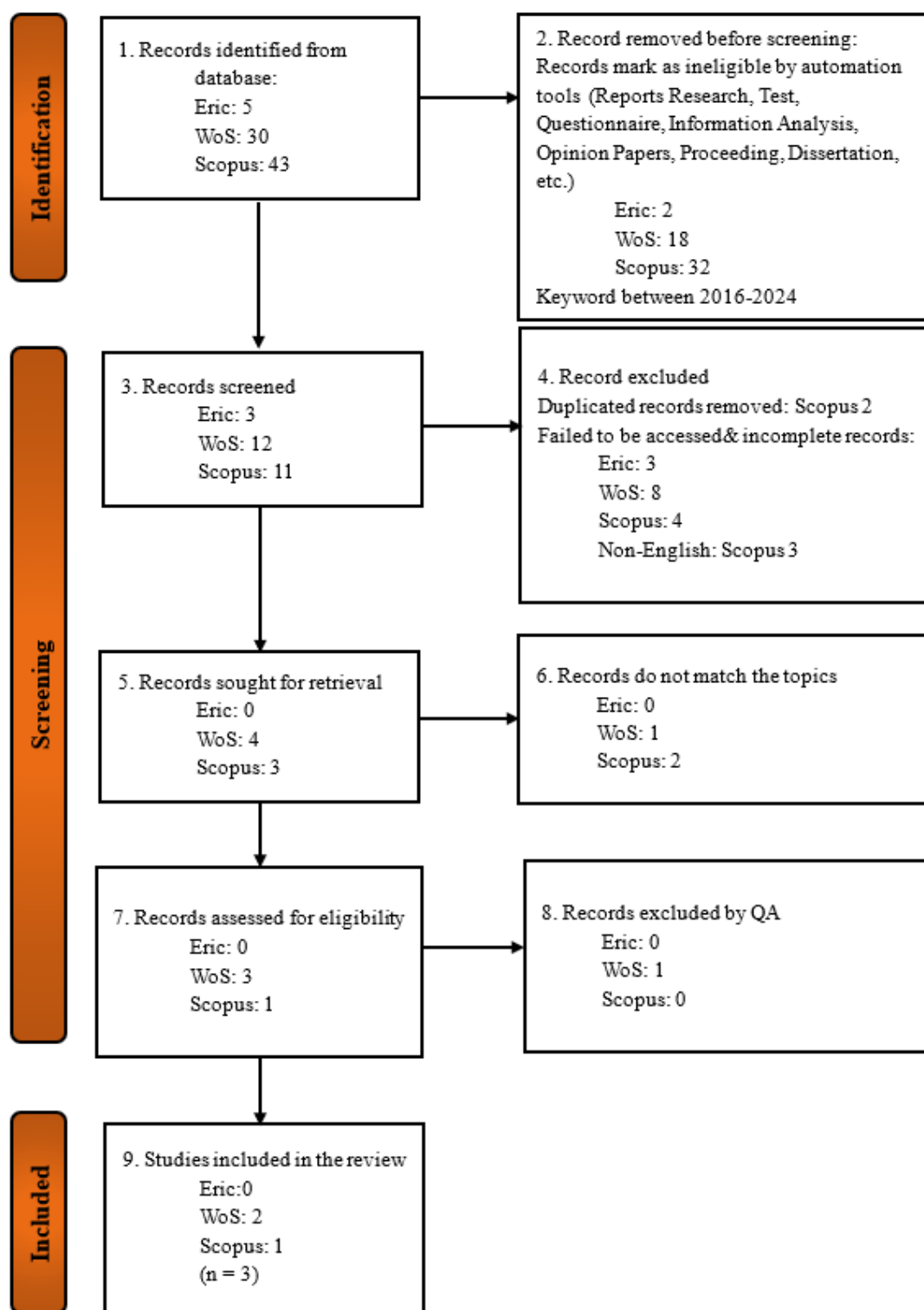


Figure 5. Result of Article Search "english for specific purposes" AND ("artificial intelligence" OR "ai")

The table presents a comprehensive synthesis of instructional strategies combining AI tools, flipped classroom techniques, CLIL-based practices, and various assessment methods used in English for Specific Purposes (ESP) instruction.

AI Tools Used

A wide range of AI tools were employed to support language learning, including:

- ChatGPT for conversation simulation, vocabulary exploration, and feedback.
- AI-based adaptive learning platforms, intelligent tutoring systems, and chatbots like Dialogflow.
- Platforms such as Classtime.com, YT Class, and Zoom enhanced interactivity.
- Other tools included paraphrasing applications, AI-based assessment platforms, and video lecture tools.

Pre-Class Activities (Flipped Learning)

- Pre-class activities focused on preparing students with foundational content, typically through:
- Self-study, micro-lesson videos, instructional YouTube links.
- Watching pre-recorded lectures, content videos, podcasts, and augmented reality games.
- Reading academic articles or content summaries before class (in some cases, submitted to peers or instructors for collaboration).

In-Class Activities (CLIL-Inspired Tasks)

- Interactive tasks dominated in-class time, including:
- Group discussions, problem-solving, role-plays, hands-on projects, and dialogues related to ESP fields like tourism or MICE.
- Students explored profession-related topics, conducted presentations, and collaborated on writing tasks.
- Activities often connected directly with pre-class materials, fostering continuity and deeper engagement.

Assessment Methods

- Assessment practices were rich and varied, combining both formative and summative approaches:
- Formative assessments included quizzes, e-portfolios, classroom observations, and ongoing feedback mechanisms.
- Summative assessments included final tests, multiple-choice exams, presentations, and writing evaluations.
- Alternative and innovative assessments involved project-based evaluations, peer reviews, reflective logs, and self-assessment tools.
- Advanced strategies such as dynamic assessment, task-based assessment, and the Occupational English Test (OET) were also mentioned.
- Several studies emphasized the importance of metacognitive awareness through tools like MARSİ and used Likert-scale surveys to measure learners' strategies and perceptions.

Table 7 summarises selected empirical studies addressing RQ1 by illustrating how flipped classroom strategies have been implemented across diverse ESP and CLIL-related contexts.

Table 7. RQ1: How are flipped classroom strategies applied in ESP and CLIL contexts?

Authors	Year	Context	Flipped Model Description	Language Focus (ESP/ CLIL)	Key Findings
Biwen Lee	2017	Taiwan	1. Pre-class video lecture 2. In-class collaborating activity	ESP: English for TOEIC test	1. The flipped classroom approach reportedly enhances students' empowerment and independent learning abilities. 2. Students reported a higher level of satisfaction with the innovative flipped classroom model
Ivana Simonova et al.	2021	Czech Republic	1. Pre-class activity using smart devices 2. In-class face- to- face activity	ESP: Academic English (AE) and English for Teaching (ET)	1. Reported differences in knowledge increase were found between coordinators and other motivation types, particularly in learning content LC2 2. Results indicated that the smart approach effectively suited students of all motivation types, enhancing their learning experience
Nicolas Montalban and Salvador Montaner- Villalba	2023	Spain	1. Hybrid approach 2. In- class project- base activity	ESP: English for Specific Purposes	1. Most questionnaire responses indicated high satisfaction and achievement levels among students 2. The hybrid didactic ESP course showed reported improvement in students' results according to the European Framework of Reference for Languages
Hao Liu et al.	2022	China	1. Out-of-class phase, content is delivered via videos for self-paced learning 2. In-class phase focuses on collaborative group or pair work	ESP: News English	1. Students' critical thinking disposition impacts their micro-lesson learning, highlighting the importance of learner attributes in flipped instruction. 2. Participants often followed teachers' lead rather than engaging in autonomous learning during the out-of-class phase
Alisa Stunžaitė et al.	2024	Lithuania	1. Study materials, such as videos and texts, before class, allowing for flexible learning 2. In-class time focuses on discussions and collaborative learning	ESP: English for Specific Purposes	1. Students reported increased satisfaction with teacher performance by semester's end. 2. Clear study materials were perceived as crucial for effective learning. 3. A shift in student attitudes indicated improved correlation between independent study material and in-class activities.
Oksana P Bykonina et al.	2019	Ukraine	1. Supports a hybrid learning environment 2. Shifts responsibility for learning from teachers to students 3. Integrates multimedia textbooks to enhance the learning experience	ESP: Business English	1. Students showed improved writing skills, response speed, and information processing abilities. 2. Tangible improvements included better test grades and positive feedback from course satisfaction surveys
Pallapa Lertcharoenwanich	2024	Thailand	1. In the 'flip' stage, students watch instructional videos to enhance problem-solving skills 2. The final 'apply' stage allows students to practice learned concepts through exercises or projects	ESP: Business English	1. Participants perceived the Explore stage as beneficial for self-study preparation and engagement 2. The Flip stage helped participants understand lessons better, though some found independent study burdensome
Mikko Hyttinen and Jarkko Suhonen	2022	Finland	1. Online Session 2. Face-to-face sessions, discussing assignments and analyzing difficult content with teacher	ESP: Basic Law	1. Students found the flipped classroom effective for promoting learning in legal courses 2. Well-designed course materials and planned classroom activities are crucial for student engagement
Ogün İlter and Savaş Yeşilyurt	2021	Turkey	1. Delivering course content via videos and supplementary materials on a learning management system (LMS)	ESP: English for Academic Pharmaceutical Purposes (EAPP)	1. Flipped learning enhances class time efficiency, differentiation, and peer interaction 2. Formative assessment in flipped learning supports scaffolding, individualization, and ongoing feedback for learners
Zheyang Xiao	2022	China	1. Teachers shift from central roles to	ESP: Business	1. The flipped learning approach reportedly improved students' translation

			facilitate a student-oriented classroom	English	competence
Oksana Bahlai et al.	2019	Ukraine	1. Students watching videos or listening to podcasts at home before in-class activities 2. In-class time was dedicated to practical applications of knowledge and developing communicative skills	ESP: Cross-cultural communication in religious tourism	2. Average scores increased in faithfulness, smoothness, and overall translation competence 1. The flipped classroom method enhances students' intercultural communicative competence and promotes self-organisation and critical thinking skills 2. The course improved students' cooperative skills and responsibility through presentations and debates
Hyung-Ji Chang	2020	Korea	1. Students prepare class content at home 2. Engage in collaborative activities during class time	ESP: Tourism English (TE)	1. Participants showed positive development in problem-solving skills, collective efficacy, and course satisfaction 2. Surveys measured problem-solving skills and collective efficacy, confirming the effectiveness of the flipped approach
Alina O Karapetian	2020	Ukraine	1. Utilizes extracurricular time for out-of-class activities 2. Class time includes discussion-based, faux-flipped, and group-based approaches	ESP: English for Specific Purposes	1. The "flipped classroom" model improved students' critical thinking skills and academic performance 2. Students in the experimental group (EG) showed an 8% improvement, while the control group (CG) improved by 1% 3. The study highlighted the importance of a student-centered approach in fostering better learning experiences
Inna Nikitova et al.	2020	Ukraine	1. Self-study assignments	ESP: English for Specific Purposes	1. Enhancing student engagement and motivation 2. A shift in student preferences towards independent learning and improved time management skills 3. The methodology could be applicable to other majors beyond language and literature teaching

Note. While the table addresses both ESP and CLIL contexts, the majority of the reviewed studies focus on ESP-specific instructional settings. CLIL-related elements are reflected in selected interdisciplinary or content-integrated language learning contexts.

Across the reviewed studies, three recurring instructional patterns were observed in the application of flipped classroom strategies within ESP and CLIL contexts (Lee, 2017; Simonova et al., 2021; Liu et al., 2022; Stunžaitė et al., 2024).

First, most studies implemented a pre-class preparation phase, in which learners engaged with video lectures, digital materials, or guided self-study resources to acquire foundational knowledge prior to classroom sessions. Second, in-class learning activities consistently emphasized interactive and task-based engagement, including group discussions, role-plays, project-based tasks, and problem-solving activities situated in professional contexts such as tourism, business, and hospitality. Third, several studies highlighted the promotion of learner autonomy and responsibility, as flipped designs encouraged students to manage their learning pace and actively apply domain-specific language during classroom interaction. Collectively, these patterns indicate that flipped classroom strategies in ESP and CLIL settings are primarily used to shift instructional focus from content delivery to applied, communicative, and profession-oriented learning experiences.

Table 8. RQ2: What roles does AI play in enhancing language learning in flipped or CLIL environments

Authors	Year	AI Tool	Learning Context	Role of AI	Impact on Learning
Christina Rapti and Panagiotidis	2024	1. Natural Language Processing (NLP) 2. Data-Driven Learning (DDL) 3. Automated Writing Evaluation (AWE) 4. Computerised Dynamic Assessment (CDA) 5. Intelligent Tutoring Systems (ITSs) 6. Automatic Speech Recognition (ASR) 7. Large Language Models (LLMs): 8. Chatbots like ELIZA, GeenieTutor, Mondly and ChatGPT	ESP teachers	1. Personalized learning experiences and efficient class administration 2. Adapt to learners' speed and style, enhancing productivity and accessibility 3. Improving engagement and learning outcomes 4. Facilitates communication and customizes learning	1. A reported number of educators utilize AI to enhance learning outcomes tailored to individual student needs
Si Wu and Fei Wang	2021	1. AI (N/A)	ESP; Listening and Speaking for English Major	1. Assist learners in solving problems encountered 2. Enhances knowledge expression ability and improves reasoning efficiency 3. Support the flipped classroom model	1. The average score of students' English self-management learning ability increased reportedly 2. Students' average professional English score improved
Harry Barton Essel et al.	2024	1.ChatGPT	EFL learners	1. Enhance students' writing skills and self-regulated learning 2. Provides personalized learning experiences 3. Facilitate critical thinking 4. Assists students in constructing knowledge	1. The study found that using ChatGPT positively influenced students' critical, reflective, and creative thinking skills
Hai-Feng Li	2023	1.MOOCs (Massive Open Online Courses) 2. English Listening Learning Platform 3. Chat GPT	EFL learners	1. Provided instant feedback and personalized scaffolding 2. Served as an adaptive scaffold library	1. Self-efficacy ratings were reportedly higher in the experimental group
Bing Li, Miaomiao Peng	2022	1. AI Language Learning Platform for English listening comprehension	EFL learners	1. Monitors and records students' behavior 2. Facilitates students' external motivation 3. Provides instant feedback and detailed learning progress	1. The study evaluated an AI-based flipped class model, revealing no reported increase in English listening scores compared to traditional methods
Ferdi Çelik and Ceylan Yangın Ersanlı	2022	1. Augmented Reality Platforms • Metaverse Studio • CoSpaces Edu • Figment AR	CLIL: ESP learners	1. Enhance learning environments	1. The experimental group showed a higher mean gain in achievement compared to the control group
Maria Tereza L. Cavalcante, Gisela Riberas, and Genoveva Rosa	2016	1. Video Creations; Moviemaker, Powtoon, 2. Content Sharing: FB 3. Learning Platforms: Moodle	CLIL: ESL learners	1. Promote problem-solving and creative thinking	1. Reported a positive perception of their competencies, with 78% scoring seven or above
Alberto Andujar, Mary Spratt	2023	1. ChatGPT 2. Language Tutoring Systems 3. Robot 4. Voice Recognition Functions 5. Translation Programs 6. Language Learning Apps	CLIL teachers	1. Support CLIL teachers 2. Help identify language competence 3. Facilitates class preparation, management, and selection of language materials	1. Psychological effects include lowered self-efficacy and fear of losing control

Kleopatra Mageira et al.	2022	1. AsasaraBot 2. AI Chatbots and Platforms; Snatchbot, DialogFlow, Rasa, OpenDialog, Freudbot, Ethnobot, Cleverbot and CSIEC, Autotutor, Chatbot (unspecified), StudyBuddy and SmarterChild, NDLtutor, Confucius, Xbot, CLIVE, Gengobot, BookBuddy system, Mondly and Andy English Bot	CLIL: ESL learners	1. Serve as intelligent tutors 2. Provide feedback 3. Support human tutors	1. Its effectiveness in teaching cultural content in foreign languages
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The synthesis of the reviewed literature indicates that Artificial Intelligence fulfills three primary pedagogical roles in flipped and CLIL-oriented language learning environments (Holmes et al., 2019; Rapti & Panagiotidis, 2024).

First, AI is frequently employed to support personalized learning and adaptive feedback, enabling learners to receive immediate responses tailored to individual proficiency levels and learning needs. Second, several studies demonstrate that AI contributes to learner autonomy and self-regulated learning, particularly by enabling independent practice through chatbots, intelligent tutoring systems, and adaptive platforms. Third, AI tools are commonly integrated as formative assessment supports, assisting instructors in monitoring learner progress and adjusting instructional strategies across pre-class and in-class phases. These findings position AI as a pedagogical enabler that enhances instructional effectiveness without replacing the central role of teachers in ESP and CLIL contexts.

Table 7 summarises reported pedagogical benefits and limitations of AI integration in ESP and CLIL classrooms based on empirical journal studies (Alharbi, 2024; Mageira et al., 2022; Andujar & Spratt, 2023).

Table 9. RQ 3: What are the pedagogical benefits and limitations of integration AI in ESP/ CLIL Classroom?

Authors	Year	Pedagogical Benefits	Pedagogical Limitations	Notes
Wael Alharbi	2024	1. Enhance the learning experience of students 2. Promote engagement and understanding 3. Personalized learning experiences through AI	1. Overreliance on AI tools may impede language skill development	1. The gap between student AI use and teacher awareness 2. Need for teacher training and ethical guidelines
Liang-Ching Chen, Kuei-Hu Chang, Hsiang-Yu Chung	2020	1. AI helps in term of ESP corpus analysis 2. Identification of Domain-Specific Language 3. Improved Accuracy and Reduced Bias	1. Challenges with Linguistic Meaning 2. Interference of Function Words 3. Manual Annotation is Time-Consuming	1. Balancing Automated Tools with Human Expertise
Oksana Synekop, Iryna Lytovchenko, Yuliana Lavrysh, and Valentyna Lukianenko	2024	1. ChatGPT enhances research skills 2. ChatGPT useful for developing both hard and soft skills in students	1. Small sample size	1. There is a need for university policies regarding AI model usage
Ferdi Çelik, Ceylan Yangın Ersanlı	2022	1. Gamification enhances student motivation and engagement 2. Augmented reality (AR) improves language skills	1. The Metaverse app had bugs on specific Android devices 2. Some students completed AR experiences with peers, limiting individual engagement	1. Non-parametric tests were employed
Maria Tereza L.et al.	2016	1. The use of CLIL aimed to enhance problem-solving and creative thinking among students	1. Challenges with Language Proficiency and Motivation	1. The work successfully combined three innovative aspects in a replicable format
Alberto Andujar, Mary Spratt	2023	1. The technology fosters skills and competences needed by teachers in CLIL contexts 2. Helps non-language teachers identify students' language competence	1. Lack of teaching resources and materials	1. CLIL teachers face challenges
Kleopatra Mageira et al.	2022	1. The use of AI chatbot technology enhances interactive ICT-based learning for foreign languages and cultural content 2. Chatbot technology allows students to learn at their own pace 3. The integration of chatbots in education can lead to positive learning outcomes	1. Chatbots struggle with recognizing natural language 2. They often misinterpret complex or unexpected questions	1. The system adheres to privacy protection policies regarding data recording

Across the reviewed studies, both pedagogical benefits and limitations of AI integration in ESP and CLIL classrooms were consistently reported (Alharbi, 2024; Mageira et al., 2022; Andujar & Spratt, 2023).

In terms of benefits, AI-supported instruction was associated with enhanced learner engagement, increased opportunities for personalized learning, and continuous formative feedback. Several studies also indicated that AI tools supported motivation and facilitated flexible, self-directed learning environments. However, the synthesis also revealed recurring limitations. These included the risk of overreliance on AI tools, technical constraints such as limited natural language understanding or system instability, and concerns related to teacher preparedness. Notably, multiple studies emphasized the importance of teacher training and institutional guidelines to ensure that AI integration remains pedagogically grounded and ethically responsible.

Table 10. RQ 4: What gaps exist in the current research landscape combining these domains?

Source	Observed Gap	Suggested Future Research
Lee (2017); Chang (2020); Liu et al. (2022); Essel et al. (2024); Mageira et al. (2022); Çelik & Yangın Ersanlı (2022)	A lack of empirical studies that holistically integrate flipped classroom pedagogy, ESP, AI, and CLIL within a single instructional design.	Design, implement, and empirically validate integrated AI-enhanced flipped CLIL models grounded in ESP needs-based pedagogy.
Ivana Simonova et al. (2021)	The research lacks a precise definition of "smart" devices	Teachers' training is required
Hao Liu, Hu Qiu, Zuocheng Zhang (2022)	There is a lack of comprehensive studies integrating various Activity systems	Explore the impact of students' critical thinking on micro-lesson learning dynamics
Biwen Lee (2017)	A notable lack of studies focusing specifically on the preparation courses	Explore more details
Alisa Stunžaitė, Pavel Skorupa, Tatjana Dubovičienė (2024)	The research highlights a gap in the widespread adoption of the flipped classroom method (FCM)	Explore more details
Oksana P. Bykonja (2019)	The research primarily focused on students majoring in Economics	Future research could involve different participants
Ogün İlter, Savaş Yeşilyurt (2021)	A gap in understanding the pedagogical dimensions	Explore action research design as a tool for professional development in flipped learning contexts
Oksana Bahlai (2019)	Its findings do not encompass the entire scope of intercultural communicative competence (ICC) development	Explore international educational experiences in developing intercultural competence through extracurricular activities
Hyung-ji Chang (2020)	The study acknowledges limitations in measuring the effectiveness of flipped classroom activities within a 15-week timeframe	Recommend to measure the effectiveness of FC activities beyond the 15-week timeframe
Inna Essel (2019)	The study's limitations include researcher bias due to involvement in course design	Future research could explore the applicability of the methodology to other majors beyond Literature and Language teaching
Christina Rapti, P. Panagiotidis (2024)	There is a lack of investigation into pedagogical applications of AI in education	Further research is likely to provide insights into AI technologies in education
Si Wu, Fei Wang (2021)	Lacks extensive exploration of its long-term effects on student performance and engagement beyond the initial experimental phase.	Future research should focus on enhancing various technologies related to flipped classrooms
Wael Alharbi (2024)	The gap in awareness between students' AI tool usage and teachers' understanding presents a reported research opportunity	Address the gap between student AI use and teacher awareness in higher education
Liang-Ching Chen, Kuei-Hu Chang, Hsiang-Yu Chung (2020)	Existing corpus software struggles with linguistic meanings	Optimize corpus analysis results and enhance efficiency in corpus-based approaches for extracting domain-oriented lexical units

Although prior studies have examined flipped ESP instruction, AI-supported language learning, or AI-enhanced CLIL environments, no empirical study identified in this review simultaneously integrates flipped classroom pedagogy, English for Specific Purposes (ESP), Artificial Intelligence (AI), and Content and Language Integrated Learning (CLIL) within a unified instructional framework.

The synthesis highlights significant gaps in the existing research landscape, particularly the lack of empirical studies that holistically integrate flipped classroom strategies, Artificial Intelligence, CLIL, and ESP within a single instructional framework (Lee, 2017; Simonova et al., 2021; Rapti & Panagiotidis, 2024).

Most reviewed studies examined these components in isolation or partial combinations, limiting insights into their combined pedagogical impact. Additional gaps include insufficient attention to teacher professional development, limited investigation of long-term learning outcomes, and a narrow focus on specific learner populations or disciplinary contexts. Several studies also noted the absence of clear conceptual definitions for AI tools and inconsistent evaluation of instructional designs. These gaps underscore the need for future empirical research that systematically designs, implements, and validates integrated AI-enhanced flipped CLIL models to advance ESP pedagogy in professional education settings.

5. Discussion

This systematic review synthesizes recent empirical evidence on the integration of Artificial Intelligence (AI), flipped classroom strategies, and Content and Language Integrated Learning (CLIL) within English for Specific Purposes (ESP) instruction, particularly in tourism and hospitality contexts. The findings indicate that AI primarily functions as a pedagogical enabler rather than a standalone technological tool, supporting personalization, formative feedback, and learner autonomy across flipped and CLIL-oriented instructional designs (Holmes et al., 2019; Rapti & Panagiotidis, 2024). Importantly, these pedagogical functions align closely with ESP needs-based pedagogy, which emphasizes aligning instructional activities, language tasks, and assessment practices with learners' professional communication needs (Dudley-Evans & St John, 1998; Anthony, 2018). From this perspective, AI-enhanced feedback and adaptive learning support mechanisms contribute to addressing the specific linguistic and situational demands encountered by ESP learners in professional contexts.

The review further demonstrates that flipped classroom strategies consistently enhance ESP learning by reallocating instructional time toward interactive, task-based, and profession-oriented activities (Bergmann & Sams, 2012; Simonova et al., 2021; Liu et al., 2022). When combined with CLIL principles, flipped learning facilitates the simultaneous development of language competence and domain-specific knowledge, which is essential for authentic professional communication in tourism and hospitality education (Coyle et al., 2010; Dalton-Puffer, 2011). Notably, the synthesis suggests that AI, flipped classroom strategies, and CLIL should not be viewed as isolated pedagogical components, but rather as interdependent elements within an instructional ecology that collectively shapes learning processes and outcomes (Luckin et al., 2016; Holmes et al., 2019). Within this ecology, ESP defines instructional relevance, CLIL structures language-content integration, flipped learning organizes temporal and spatial learning activities, and AI enhances instructional responsiveness and adaptability.

Despite these advances, the synthesis reveals a limited number of studies that holistically integrate all four dimensions of AI, flipped classrooms, CLIL, and ESP within a single instructional framework. Most existing research addresses these components in partial combinations, underscoring the need for future empirical studies that systematically design and validate integrated AI-enhanced flipped CLIL models grounded in ESP needs-based pedagogy (Coyle et al., 2010; Rapti & Panagiotidis, 2024).

6. Conclusion

This systematic review synthesized empirical studies published between 2016 and 2024 that examined the integration of Artificial Intelligence (AI), flipped classroom strategies, and Content and Language Integrated Learning (CLIL) in English for Specific Purposes (ESP) instruction, with a particular focus on tourism and hospitality contexts. The findings indicate that each component contributes complementary pedagogical value: ESP provides a needs-based foundation that aligns language instruction with professional communication demands (Dudley-Evans & St John, 1998; Anthony, 2018), CLIL facilitates the integration of language and domain-specific knowledge (Coyle et al., 2010), flipped classroom strategies promote interactive and task-based learning (Bergmann & Sams, 2012), and AI enhances personalization, formative feedback, and learner autonomy (Holmes et al., 2019).

Importantly, the synthesis suggests that these elements are most effective when conceptualized as an instructional ecology, in which AI, flipped learning, and CLIL function as interdependent pedagogical mechanisms rather than isolated interventions (Luckin et al., 2016). Despite promising findings, the review reveals a limited number of studies that holistically integrate all four dimensions within a single instructional framework. Future research should therefore focus on the systematic design, implementation, and empirical validation of AI-enhanced flipped CLIL models grounded in ESP needs-based pedagogy to advance professional language education in technology-rich learning environments.

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Authors contributions

Weerapa Pongpanich, the first author, conceptualized the research framework, conducted the systematic literature review, performed data extraction and synthesis, and prepared the initial draft of the manuscript.

Assistant Professor Dr. Thada Jantakoon, the principal advisor, supervised the validation process and provided comprehensive academic guidance throughout the manuscript's review and revision stages.

Assistant Professor Dr. Rukthin Laoha, the second advisor, reviewed the manuscript and provided constructive scholarly feedback to enhance its academic quality and methodological precision. All authors have read and approved the final version of the manuscript.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Obtained.

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The Publication Ethics Committee of the Canadian Center of Science and Education. The journal's policies adhere to the Core Practices established by the Committee on Publication Ethics (COPE).

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Appendix A

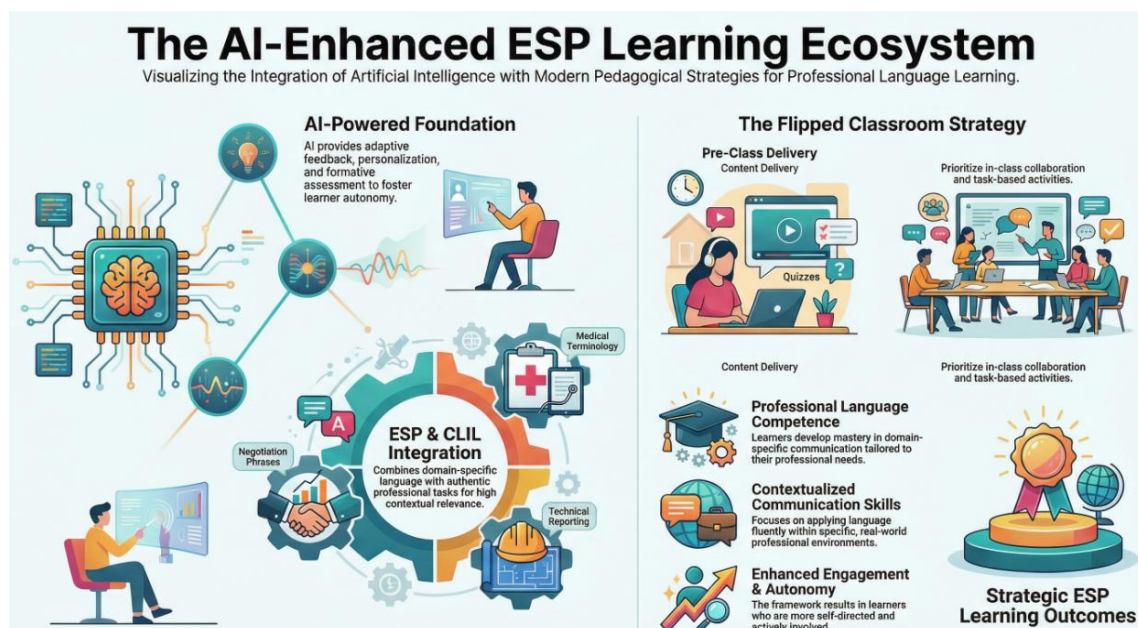


Figure A1. Synthesized Conceptual Representation of the AI-Enhanced ESP Learning Ecosystem

Figure A1 presents a synthesized conceptual representation of the AI-enhanced English for Specific Purposes (ESP) learning ecosystem, derived from a systematic synthesis of the empirical studies included in this review. The figure does not propose a new instructional model; rather, it consolidates recurring pedagogical patterns, instructional roles, and structural relationships identified across the reviewed literature.



Figure A2. Synthesized Instructional Ecology of ESP Integrating AI, Flipped Learning, and CLIL

Figure A2 presents a synthesized instructional ecology of English for Specific Purposes (ESP) derived from the systematic synthesis of reviewed empirical studies. The figure does not introduce a new instructional model; rather, it visualizes recurring pedagogical roles, learning structures, and instructional phases consistently reported in the literature. ESP is positioned as the pedagogical foundation defining professional language objectives, CLIL functions as the integration mechanism linking language and domain-specific knowledge, and flipped learning operates as the structural strategy organizing pre-class, in-class, and assessment activities. Artificial Intelligence is depicted as a pedagogical enabler supporting personalization, learner autonomy, and formative assessment across all phases. Overall, the figure provides an integrative conceptual lens grounded in existing empirical evidence.